

Statistics - Set 1 - Solutions

Specification M6.1-M6.4.

Answer key

Question	Answer	Topic
1	A	Statistics
2	C	Statistics
3	C	Statistics
4	F	Statistics
5	F	Statistics
6	A	Statistics
7	D	Statistics
8	F	Statistics
9	D	Statistics
10	D	Statistics
11	H	Statistics

Worked solutions

Question 1 STA-01 · Statistics · spec M6.3

The mean of five numbers is 12. When a sixth number is added the mean becomes 14. What is the sixth number?

Answer: A

Total goes from 60 to 84: the sixth number is 24.

Question 2 STA-02 · Statistics · spec M6.3

A histogram has a bar for the class $10 \leq x < 15$ of height (frequency density) 3.2 and a bar for $15 \leq x < 25$ of height 1.5. What is the total frequency of these two classes?

Answer: C

Frequency = density $\times$ width: $3.2 \times 5 + 1.5 \times 10 = 16 + 15 = 31$.

Question 3 STA-03 · Statistics · spec M6.3

The numbers 3, 7, x , 12, 18 (with $7 \leq x \leq 12$) have mean equal to median. Find x .

Answer: C

The median is x and the mean is $\frac{40+x}{5}$. Setting them equal: $40 + x = 5x$, so $x = 10$.

Question 4 STA-04 · Statistics · spec M6.3

The table shows grouped data: $0 \leq t < 10$: frequency 4; $10 \leq t < 20$: frequency 10; $20 \leq t < 40$: frequency 6. Estimate the mean.

Answer: F

Use the midpoints 5, 15, 30: $\frac{4 \cdot 5 + 10 \cdot 15 + 6 \cdot 30}{20} = \frac{20 + 150 + 180}{20} = \frac{350}{20} = 17.5$.

Question 5 STA-05 · Statistics · spec M6.4

Which of the following statements about a scatter graph showing strong positive correlation between hours of revision and exam mark is/are valid?

- I Students who revised more tended to score higher.
- II Revising more causes a higher mark.
- III A line of best fit could be used to estimate the mark of a student who revised for a number of hours within the range of the data.

Answer: F

Correlation describes a tendency (I) and supports interpolation within the data range (III), but does not by itself establish causation (II).

Question 6 STA-06 · Statistics · spec M6.3

A set of n numbers has mean m . Each number is doubled and then 3 is added. Which of the following is the new mean, and which the new range (if the original range was R)?

Answer: A

Mean transforms with the data: $2m + 3$. The range is unaffected by adding a constant but doubles with the scaling: $2R$.

Question 7 STA-08 · Statistics · spec M6.3

A cumulative frequency graph for 80 observations passes through (20, 20), (30, 40) and (45, 60). Estimate the interquartile range.

Answer: D

Lower quartile at cumulative frequency 20: value 20. Upper quartile at 60: value 45. IQR = 25.

Question 8 STA-09 · Statistics · spec M6.3

Class A has 20 students with mean mark 60; class B has 30 students with mean mark 70. What is the mean mark of all 50 students?

Answer: F

$$\frac{1200 + 2100}{50} = \frac{3300}{50} = 66.$$

Question 9 STA-10 · Statistics · spec M6.3

For which of the following data sets is the mean greater than the median?

Answer: D

A large outlier pulls the mean above the median: mean 6 > median 3. In B, D, E the mean equals the median; in C the mean 9.4 < median 11.

Question 10 STA-07 · Statistics · spec M6.3

The median of 7 distinct integers is 10 and the mean is 12. The smallest number is increased by 20. Which of the following must be true?

Answer: D

The mean rises by $\frac{20}{7}$. The smallest number plus 20 could exceed the median (e.g. if the smallest is 1 and it becomes 21 the middle value changes), or it could stay below it. So the median may or may not change.

Question 11 X2-28 · Statistics · spec M6.3

A set of five distinct positive integers has median 10 and mean 10. Which of the following statements must be true?

- I The largest value is at least 12.
- II The smallest value is at most 8.
- III The range is at least 4.

Answer: H

The two values above the median are distinct integers greater than 10, so at least 11 and 12 (I); similarly the two below are at most 9 and 8 (II); hence the range is at least $12 - 8 = 4$ (III).